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A randomised pilot trial for bedtime procrastination: Examining the efficacy and feasibility of the Reducing Evening Screen Time online intervention (REST-O)☆

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ABSTRACT

Bedtime procrastination is associated with inadequate sleep and is a novel intervention target to improve sleep health. Formative work indicates that pre-sleep electronic device use, a ubiquitous bedtime procrastination behaviour, may be targeted using a behaviour change approach. This pilot study aimed to examine (1) the acceptability and feasibility of the Reducing Evening Screen Time online program (REST-O) in new career starters, and (2) the preliminary effect of the program on daily pre-sleep electronic device use and sleep duration, as well as weekly measures of bedtime procrastination, excessive device use, and sleep quality and duration. Participants (N = 55) were randomised into three arms; an active Control (n = 19), Prevent (n = 18) and Substitute (n = 18), who used behavioural substitution at different times of day. Daily assessments were conducted for two weeks, and weekly measures at baseline, pre-intervention, post-intervention, and follow-up (end of Week Three). All groups experienced a reduction in daily pre-sleep device use (M = 23.4 min per day) and an increase in daily sleep duration (M = 12.7 min per day) post-intervention. Continued reductions in bedtime procrastination, excessive device use, and improvements in sleep quality and duration were observed at follow-up. The program appears feasible and acceptable to participants with a high daily completion rate (M = 84.7 %). Reductions in habit strength and hedonic motivation suggest potential mechanisms of change. Findings provide preliminary evidence for behaviour change interventions in targeting pre-sleep electronic device use and bedtime procrastination, with broader implications for sleep health.

1. Introduction

A contributor to inadequate sleep, bedtime procrastination is the volitional delay of bedtime without the influence of external factors (such as disturbances from work or children), where one expects to be worse off because of the delay [1]. Reasons underpinning bedtime procrastination include a lack of self-control, not feeling tired, or a desire for entertainment [2], and the phenomenon is linked to negative health outcomes: bedtime procrastination is associated with decreased sleep quality and sleep duration, and increased daytime fatigue [1,3]. As more than half of adults report procrastinating bedtime two or more nights a week [4], bedtime procrastination may be a modifiable health behaviour to help alleviate the burden of inadequate sleep.

Bedtime procrastination can encompass many activities; however, qualitative reports almost exclusively detail electronic device use as the behaviour used to procrastinate bedtime [2]. Electronic device use describes using smartphones, tablets, gaming consoles, computers, and televisions to facilitate exposure to media, information, and communication in the pre-sleep period ([5]; *also referred to as screen use,* [6]; *screen time,* [7]; *and electronic media use,* [8]). Pre-sleep electronic device use is ubiquitous: 9 out of 10 people report using an electronic device in the hour before bedtime [9,10] and 83 % have smartphones within reach from bed [11]. Electronic device use immediately before or in-bed has been associated with general sleep disturbances [12], including

☆ This work is not under review elsewhere and has not been published before.

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delayed sleep onset, decreased sleep quality, and shorter sleep duration [5,13,14]; and as such, has been the focus of intervention efforts.

To date, the majority of interventions to reduce electronic device use have targeted adolescent populations, largely focusing on device restriction measures such as digital curfews [15–17], device avoidance [18] and parental involvement [19]. Few interventions have targeted adult populations and the pre-sleep period [20], particularly emerging adults who cite newfound freedom living alone and a lack of accountability as factors influencing pre-sleep electronic device use [2]. Due to the ubiquity of electronic devices, device restriction may not be effective in reducing use in adults over the long-term: intervention approaches based on willpower often lead to a decline back to baseline behaviour [21]. Further, despite the link between pre-sleep electronic device use and inadequate sleep [5,13,14], people report not wanting to give up their devices or me-time [2]. Therefore, a more practical approach may be to focus on disrupting or reducing time spent using electronic devices in the pre-sleep period to form new habitual behaviours, rather than relying on device restriction or willpower, which may be ineffective in the long term. To achieve this, intervening with the reward pathways/motivation systems that underlie engagement with electronic devices is promising.

Our formative work suggests that dual cognitive processes and hedonic motivation underlie pre-sleep electronic device use ([2]; Hill et al., *under review)*. Dual process theories account for automatic processing (habitual, impulsive, non-conscious responses) and reflective processing (reasoned, conscious decision making) [22]. Pre-sleep electronic device use is described in terms of both processes: it can be initiated mindlessly or involve internal debate about the consequences of procrastinating sleep [2]. However, interventions targeting reflective processes have not been successful in reducing bedtime procrastination [23]. In contrast, recent interventions with a behavioural component have been effective in reducing bedtime procrastination in a general population [24]. Thus, focusing on automatic processes may be a more promising approach for changing bedtime procrastination behaviours [25]. Further, formative work indicated that hedonic motivation, a form of automatic processing where impulsive reactions to behavioural cues are based on past pleasant or unpleasant experiences [26], underlies pre-sleep behaviour. Individuals with stronger hedonic impulses to engage in pre-sleep electronic device use are more likely to use devices, both on a daily basis and overall (Hill et al., *under review*). Therefore, interventions that target the point of initiation of impulsive behaviours and pleasant substitute behaviours could be effective in reducing pre-sleep electronic device use. Behavioural substitution, a Behaviour Change Technique (BCT) where an unwanted behaviour is replaced with a wanted behaviour [27], can disrupt learned associations between behavioural cues, initiation, and rewards [28], and may be a promising in modifying pre-sleep behaviours.

We developed a theory-driven, behaviour change intervention aimed at reducing pre-sleep electronic device use in new career starters, the Reducing Evening Screen Time online program (REST-O). New career starters, those transitioning from tertiary education to the workforce, may be at risk of inadequate sleep as recent graduates [29] and are experiencing a "fresh start", an opportune time to change behaviour [30]. Through a three-arm randomised design with an active control utilising goal setting (Control), we investigated the effects of two habit-based approaches to reduce bedtime device use: where a substitute behaviour is adopted in the pre-sleep period (Substitute), or where a substitute behaviour is adopted in the daytime to prevent pre-sleep electronic device use (Prevent). As the comparison between the daytime and pre-sleep period is exploratory, there is no specific hypothesis regarding the intervention group difference.

The aim of the present study is to assess the efficacy and feasibility of an intervention for pre-sleep electronic device use in new career starters. The primary outcomes were daily pre-sleep electronic device use and daily sleep duration. The secondary outcomes were bedtime procrastination, excessive electronic device use, and weekly sleep quality and duration. Habit strength and hedonic motivation toward pre-sleep electronic device use were measured as mechanisms of change. Feasibility items included daily survey completion rate, intervention attrition rate, and the effect of the intervention on outcomes. It was hypothesised that:

(1) The Substitute and Prevent groups will experience a decrease in daily pre-sleep electronic device use post-intervention (in Week Two) compared with Control group.
(2) The Substitute and Prevent groups will experience improved daily sleep duration post intervention (in Week Two compared to Week One).
(3) The Substitute and Prevent groups will have decreased bedtime procrastination scores, decreased excessive device use scores, and improved sleep quality and duration at follow-up compared with baseline.

2. Methods

2.1. Intervention design

The intervention design is presented in Fig. 1, including BCTs identified from the Theory and Techniques Tool (TATT) [31].

The duration of the REST-O program was 14 days, with the core intervention component (i.e., informational video, problem-solving coaching session and substitute behaviour or goal setting adoption) occurring on days 6–7 (end of Week One). Final follow-up measures were conducted two weeks later at the end of Week Three. Delivery was primarily by text messages, or email when requested by participants. For the first 7 days (Week One), participants self-monitored feelings (BCT 5.4), duration of key behaviours (BCT 2.3), and physiological outcomes of behaviour (e.g., daytime fatigue; BCT 2.4) using a daily sleep diary. At the end of Week One, a 15-min informational video outlined BCTs, covering either goal setting (Control; BCT 1.1) or training on habit cues, rewards and behavioural substitution (Prevent and Substitute; BCT 8.2). A screen shot from the video is presented in Supplemental Figure 1. Following the video, participants attended a personalised 15-min problem-solving coaching session on Zoom [32] to set generic goals (Control) or to problem-solve to determine substitute behaviour (BCT 1.2) and set goals accordingly (Substitute/Prevent). For the following 7 days (Week Two), participants self-monitored behaviour and completed a daily sleep diary. Weekly surveys were filled out at baseline, end of Week One (pre-intervention), end of Week Two (post-intervention), and end of Week Three (follow-up). A small completion incentive, a A$20 gift card, was offered on completion of the follow-up survey. Further detail regarding the needs assessment, intervention development, and testing is presented in the Supplemental materials. Ethical approval was obtained from CQUniversity Human Research Ethics Committee (Approval 24,611), and the study was prospectively registered with the Australian New Zealand Clinical Trials Registry (ACTRN12624000105549).

2.2. Participants

Participants were recruited from social media and gave written informed consent. Participants were screened against inclusion criteria to ensure they (1) had finished full-time tertiary study and started full-time employment in the past 12 months, (2) were 18 years of age or older, and (3) engaged in bedtime procrastination. Those previously treated for a sleep disorder were excluded to minimise confounding factors.

2.3. Randomisation

Participants were randomised into 3 groups in a 1:1:1 ratio upon signup, utilising block randomisation in a random sequence (e.g., CSP,

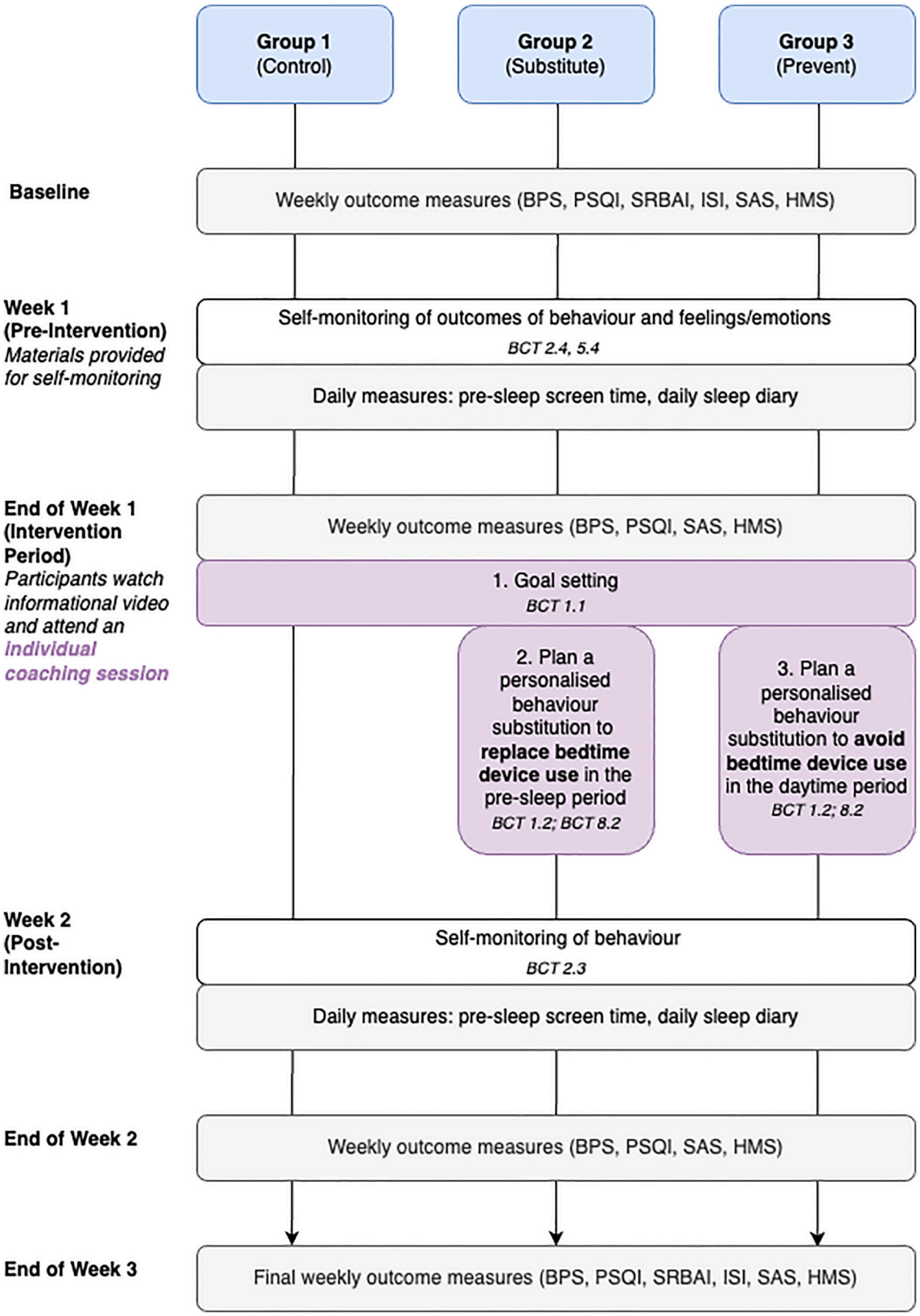


Fig. 1. *Intervention design using Behavioural Change Techniques (BCTs) from the Theory and Techniques Tool (TATT).*
Note. BPS = Bedtime Procrastination Scale, PSQI = Pittsburgh Sleep Quality Index, SRBAI = Self-Report Behavioural Automaticity Index, ISI = Insomnia Severity Index, SAS = Smartphone Addiction Scale, HMS = Hedonic Motivation Scale, BCT = Behaviour Change Technique.

PCS, SPC, PSC, SCP, CPS; where C = Control, S = Substitute, P = Prevent) and were blind to their assigned group [33].

2.4. Baseline, weekly and follow-up measures

2.4.1. Bedtime procrastination

The Bedtime Procrastination Scale (BPS) [1] assessed bedtime procrastination. Participants reported frequency with which they agreed with 9 items during the past week on scales ranging from 1 (*never*) to 5 (*always*). Interitem reliability was high with Cronbach's α = 0.87 and Guttman's λ_6 = 0.88.

2.4.2. Sleep quality

Sleep quality was assessed with the 6-item version of the Pittsburgh Sleep Quality Index (B-PSQI) [34]. Participants reported average sleep onset latency, bedtime, rise time, sleep duration, nighttime awakenings and sleep quality during the past week. Both Cronbach's α and Guttman's λ_6 = 0.58. Cronbach's α rose to 0.64 and Gutman's λ_6 to 0.6 when the item measuring nighttime awakenings was removed.

2.4.3. Habit strength

Habit strength of bedtime device use was measured with the Self-Report Behavioural Automaticity Index (SRBAI) [35]. Participants respond on a scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*) across 4-items (e.g., "Thinking about using your electronic device to procrastinate sleep … I did it automatically") (Cronbach's α = 0.91, Guttman's λ_6 = 0.89).

2.4.4. Excessive device use

The 10-item short version of the Smartphone Addiction Scale (SAS-SV) [36] was used to assess repetitive and excessive device use. Although this study examined electronic device use more broadly, the SAS-SV was selected due to its ability to capture key aspects of problematic engagement with devices, including cognitive preoccupation and dependency. Participants agreed with statements indicative of problematic usage during the past week on a scale from 1 (*strongly disagree*) to 6 (*strongly agree*) (Cronbach's α = 0.79, Guttman's λ_6 = 0.82).

2.4.5. Motivation

Hedonic and reflective motivation for pre-sleep electronic device use were measured using 2-items of the 4-item Hedonic Motivation Scale (HMS) (Hill et al., *under review*). As each HMS item assesses a single construct, the remaining two items measuring hedonic and reflective motivation to sleep were not relevant to this study; however, their exclusion does not impact the reliability of the scale, as each construct is measured with a single item. Questions captured hedonic desire for pre-sleep electronic device use ("Did you have an urge to use your device?") and reflective motivation for pre-sleep electronic device use ("Did you feel obligated to use your device for work or other commitments?"). Participants responded from 0 (*not at all*) to 10 (*very much*) on a visual analogue scale. As each construct is measured with a single item, a reliability coefficient was not appropriate [37] and a histogram was visually inspected to ensure normal distribution.

2.4.6. Insomnia

The Insomnia Severity Index (ISI) [38] assessed insomnia symptoms: although not an outcome measure or exclusion criteria, bedtime procrastination and insomnia symptoms share numerous characteristics (Hill et al., *under review*). In the present sample, M = 11.36 (SD = 5.36), not indicative of clinical insomnia symptoms (Cronbach's α = 0.79 and Guttman's λ_6 = 0.80).

2.5. Daily measures

2.5.1. Sleep diary

Each morning, participants logged the previous evening's procrastinatory pre-sleep electronic device use, sleep onset time, and current wake time [39]. Pre-sleep electronic device use was estimated for the previous evening and measured in minutes, utilising the same approach as our formative work (Hill et al., *under review)*.

2.5.2. Self-monitoring

Each morning, participants were prompted to write three open responses monitoring outcomes of behaviour and feelings (e.g., "Thinking about procrastinating your bedtime with your device last night … Why did you start procrastinating?").

2.6. Data analysis

Data analysis was conducted in R version 4.2.1 [40], using lme4 [41] and emmeans [42] packages. Participant characteristics were summarised using means and standard deviations. To evaluate feasibility, we calculated retention rates from allocation to follow-up, as well as daily survey completion and adherence to goals/substitute behaviours; and reported as percentages or mean days where appropriate. During the coaching session, participants' goals, substitute behaviours, and open text responses addressing feasibility and acceptability were recorded verbatim by researchers.

We conducted assumption testing by inspections of visual distribution of plots and descriptive statistics [41]. To test hypotheses, we fit linear mixed models with group and week as fixed effects, and subject (i.e., participant) as the random intercept to account for repeated measures within-person across weeks and to accommodate missingness. For group and week comparisons, estimated marginal means were calculated and reported, providing means, standard errors, and 95 % confidence intervals (95 % CI). The 95 % CI and p-values are reported alongside the estimated marginal means; however, these are not the primary focus of interpretation. Given the pilot nature, the study is (intentionally) underpowered for generalisation; rather, the findings are intended to guide proof-of-concept estimations of effect sizes for future research. Therefore, interpretations emphasise effect sizes, with Cohen's $d \geq 0.3$ considered a moderate effect and Cohen's $d \geq 0.8$ a large effect [43].

3. Results

3.1. Sample description

The study was conducted from April to June 2024 inclusive of recruitment and follow-up. Ninety-seven participants responded to recruitment posts, 81 were eligible to begin Week One assessments. Of the participants who were not eligible, 12 did not complete the baseline measures and four were excluded for data integrity concerns for shared IP addresses and open-response answers, potentially claiming multiple gift card incentives. Eighty-one participants were randomised. Of these participants, 10 did not complete one daily diary entry in Week One, 15 did not schedule their coaching session, and one did not complete one daily diary entry in Week Two. The final sample consisted of 55 participants across the Control (n = 19), Prevent (n = 18), and Substitute (n = 18) groups. The consolidated standards of reporting trials flow diagram (CONSORT; [44]) is presented in Fig. 2.

Participants had a mean age of 28.7 years (SD = 6.5) and were 64 % male (n = 36), 25 % female (n = 14), 7 % non-binary (n = 4), and 4 % gender fluid (n = 2). Participants had been employed an average of 6.4 months (SD = 5.7) and most commonly had a bachelors (54 %, n = 30) or graduate degree (30 %, n = 17), or had completed some university or technical school (9 %, n = 5). Participants resided internationally, in the United States of America (n = 16), Australia (n = 15), Canada (n = 7), Central and South America (n = 5), continental Europe (n = 5), United

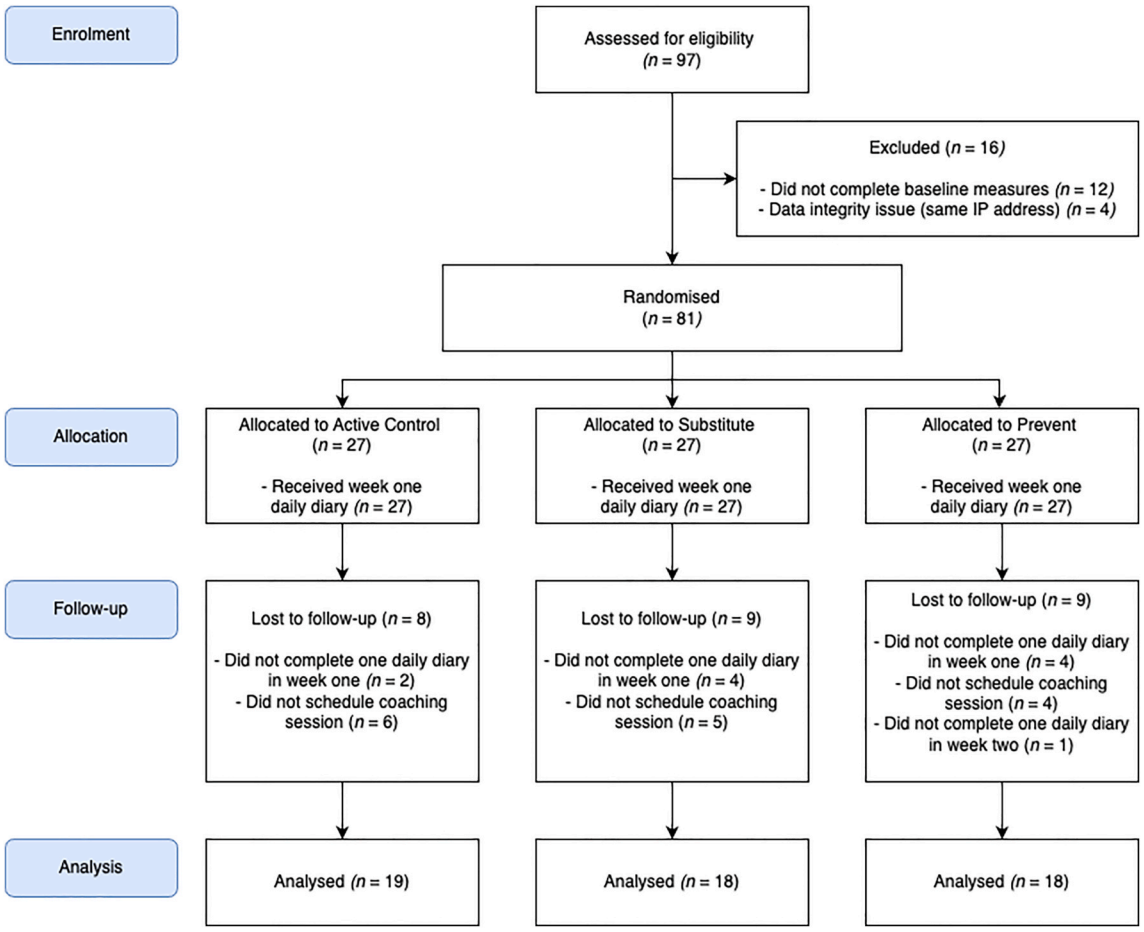


Fig. 2. CONSORT flow diagram for the present study.

Kingdom ($n = 3$), India ($n = 2$) and Africa ($n = 2$). Goals and substitute behaviours adopted by participants are presented in Supplemental Table 1.

3.2. Primary outcomes: pre-sleep electronic device use and sleep duration

The pre-post intervention changes for daily pre-sleep electronic device use are presented in Table 1. All groups experienced a decrease in pre-sleep electronic device use from pre-to post-intervention. In support of hypothesis one, there were significant decreases in pre-sleep electronic device use in the Prevent ($M = 28.7$ min/day, $SE = 12.2$, $p = 0.02$) and Substitute groups (23.1 min/day, $SE = 11.5$ $p = 0.04$), both with a moderate effect size. Further, the Control group saw an average decrease of 18.4 min/day ($SE = 11.5$ $p = 0.11$), also with a moderate effect size. There were no statistically significant differences between groups ($p > 0.05$). All pre-post intervention group changes in pre-sleep electronic device use are visualised in Fig. 3.

The pre-post intervention changes in daily sleep duration are displayed in Table 1. All groups experienced small increases in mean daily sleep duration. Daily sleep duration saw a slight, non-significant increase in both the Prevent group ($M = 8.8$ min/day, $SE = 18.2$, $p = 0.63$) and Substitute group ($M = 4.8$ min/day, $SE = 17.3$, $p = 0.78$) post-intervention (during Week Two), with a small effect size. As the increase was non-significant, hypothesis two was not supported. Additionally, the Control group saw a non-significant increase in daily sleep duration post-intervention ($M = 24.2$ min/day, $SE = 17.2$, $p = 0.18$), with a small effect size. There were no statistically significant group differences ($p > 0.05$). All group changes are visualised in Fig. 4.

Table 1
Estimated marginal means for daily outcome variables.

	Group	*N*	Pre-Intervention (Week 1)		95 % CI		Post-Intervention (Week 2)		95 % CI		Change (Week 1-Week 2)		Effect Size (Week differences), *p*
			M	*SE*	Lower	Upper	*M*	*SE*	Lower	Upper	*M*	*SE*	*d (p-value)*
Daily pre-sleep electronic device use (mins)	Control	19	47.6	8.1	31.7	63.5	29.2	8.2	13.2	45.1	18.4	11.5	−0.4 (0.11)
	Prevent	18	63.0	8.6	46.1	79.9	34.4	8.6	17.5	51.2	28.7	12.2	−0.6 (0.02)
	Substitute	18	43.8	8.1	28.0	59.7	20.7	8.2	4.6	36.8	23.1	11.5	−0.5 (0.04)
Daily sleep duration (h:mm)	Control	19	7:09	12.1	6:45	7:33	7:33	12.2	7:09	7:57	−0:24	17.2	0.2 (0.16)
	Prevent	18	7:24	12.9	6:58	7:49	7:33	12.9	7:07	7:58	−0:09	18.2	0.1 (0.63)
	Substitute	18	7:21	12.0	6:58	7:45	7:26	12.5	7:01	7:50	−0:05	17.3	0.1 (0.78)

Note. There were no statistically significant between-group differences at $p < 0.05$. Significant differences between weeks within-group are highlighted in bold text.

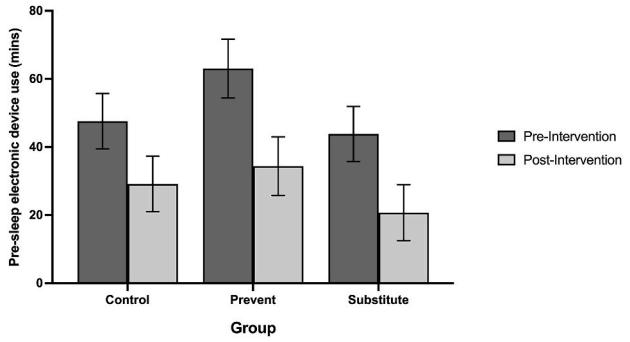


Fig. 3. Pre-post intervention daily pre-sleep electronic device use across groups.

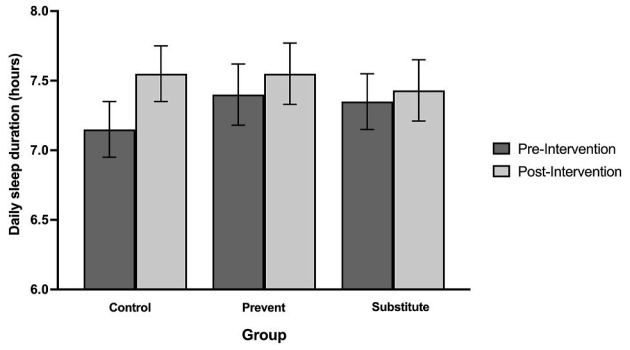


Fig. 4. Pre-post intervention daily sleep duration across groups.

3.3. Primary Outcomes: weekly bedtime procrastination, excessive device use, sleep quality and duration

Changes in outcome variables assessed on a weekly basis – bedtime procrastination, excessive device use, sleep quality and duration – are presented in Table 2. Variables were assessed at baseline, end of Week One (pre-intervention), end of Week Two (post-intervention) and end of Week Three (follow-up). In partial support of hypothesis three, the Substitute and Prevent groups reported significantly lower bedtime procrastination at follow-up compared with baseline, both with a large effect size. The Prevent group also reported significantly lower excessive device use at follow-up compared with baseline, with a moderate effect size. There were no statistically significant between-group differences for any week across all models ($p > 0.05$). Week-by-week changes for all groups are visualised in Fig. 5.

3.4. Secondary outcomes: habit strength and hedonic motivation

Change in habit strength and hedonic motivation towards pre-sleep electronic device use is presented in Table 3. The Control, Substitute, and Prevent groups all saw a significant decrease in habit strength for pre-sleep electronic device use from baseline to follow-up, all with a large effect size. Similarly, the Substitute, Prevent, and Control groups all experienced a significant decline in hedonic motivation for pre-sleep electronic device use from baseline to follow-up, with a large effect size. There were no significant between-group differences in either habit strength or hedonic motivation (all $p > 0.05$). Fig. 6 displays the mean baseline, pre-post intervention and follow-up changes in hedonic motivation for pre-sleep electronic device use.

Table 2
Estimated marginal means for weekly variables.

	Group	Baseline	95 % CI		End of Week 1	95 % CI		End of Week 2	95 % CI		Follow-up (End of Week 3)	95 % CI		Effect size
		M (SE)	Lower	Upper	*M*	Lower	Upper	*M*	Lower	Upper	*M*	Lower	Upper	*d*
Bedtime procrastination (BPS)	Control	3.6 (0.2)	3.3	3.9	3.7 (0.1)	3.4	4.0	3.2 (0.1)	2.9	3.4	3.0 (0.2)	2.7	3.3	−0.7
	Prevent	**3.8 (0.2)**	**3.4**	**4.1**	3.9 (0.1)	3.6	4.2	3.2 (0.1)	2.9	3.5	**3.0 (0.2)**	**2.7**	**3.3**	−1.0
	Substitute	**3.8 (0.2)**	**3.5**	**4.1**	3.7 (0.1)	3.4	4.0	3.1 (0.1)	2.9	3.4	**3.0 (0.2)**	**2.7**	**3.3**	−1.4
Excessive device use (SAS-SV)	Control	3.9 (0.4)	3.1	4.6	3.7 (0.3)	3.0	4.3	3.29 (0.3)	2.6	4.0	2.8 (0.4)	2.0	3.6	−0.6
	Prevent	**5.0 (0.4)**	**4.2**	**5.8**	5.0 (0.4)	4.3	5.7	4.13 (0.4)	3.5	4.8	**3.3 (0.4)**	**2.5**	**4.1**	−0.7
	Substitute	5.0 (0.4)	4.3	5.7	4.7 (0.3)	4.1	5.4	3.84 (0.3)	3.2	4.5	3.7 (0.4)	3.0	4.5	−1.1
Sleep quality (PSQI)	Control	7.2 (0.7)	5.9	8.5	6.0 (0.6)	4.8	7.2	5.5 (0.6)	4.3	6.7	5.1 (0.7)	3.78	6.4	−0.6
	Prevent	7.0 (0.7)	5.2	8.0	6.3 (0.6)	5.0	7.5	5.1 (0.6)	3.9	6.4	4.0 (0.7)	2.59	5.33	−1.1
	Substitute	8.1 (0.7)	6.8	9.4	7.7 (0.6)	6.5	8.9	6.5 (0.6)	5.3	7.7	6.2 (0.7)	4.89	7.48	−0.7
Sleep duration (h:mm)	Control	6:23 (14.9)	5:55	6:51	6:37 (13.0)	6:11	7:02	6:46 (13.0)	6:20	7:11	6:53 (14.4)	6:25	7:21	0.4
	Prevent	6:32 (14.9)	6:03	7:01	6:52 (13.7)	6:25	7:19	6:49 (13.7)	6:22	7:16	7:21 (15.0)	6:52	7:51	0.9
	Substitute	6:35 (14.1)	6:08	7:03	6:47 (12.9)	6:22	7:12	6:53 (13.0)	6:27	7:18	7:00 (14.2)	6:32	7:28	0.6

Note. Significant differences between weeks within-group are highlighted in bold text. There were no statistically significant between-group differences at $p < 0.05$. The bedtime procrastination scale (BPS) is measured from 1 to 5 and the smartphone addiction scale (SAS) 1–10, with higher scores representing higher bedtime procrastination and excessive device use. The Pittsburgh sleep quality index (PSQI) is measured from 0 to 15, with higher scores indicating lower sleep quality.

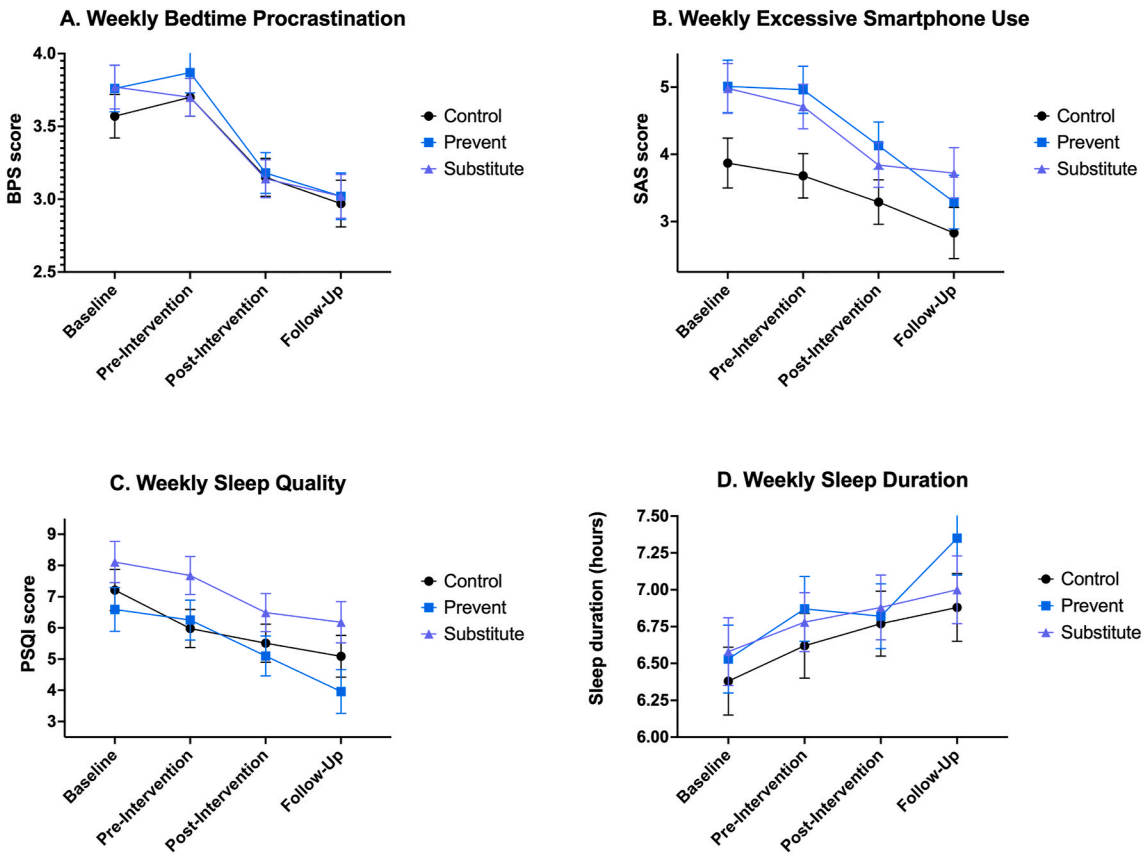


Fig. 5. Weekly changes in bedtime procrastination (A), excessive device use (B), sleep quality (C), and sleep duration (D).
Note. BPS = Bedtime Procrastination Scale, SAS = Smartphone Addiction Scale, PSQI = Pittsburgh Sleep Quality Index. Lower BPS and SAS scores indicate reduced levels of these outcomes, lower PSQI scores indicates higher sleep quality.

Table 3
Estimated marginal means for secondary outcomes.

	Group	Baseline	95 % CI		End of Week 1	95 % CI		End of Week 2	95 % CI		Follow-up (End of Week 3)	95 % CI		Effect size
		M (SE)	Lower	Upper							*M*	Lower	Upper	*d*
Habit strength	Control	5.1 (0.4)	**4.4**	**5.8**	–	–	–	–	–	–	3.5 (0.4)	**2.7**	**4.2**	**−0.8**
	Prevent	5.8 (0.4)	**5.1**	**6.5**	–	–	–	–	–	–	4.0 (0.4)	**3.3**	**4.7**	**−1.2**
	Substitute	5.5 (0.4)	**4.9**	**6.2**	–	–	–	–	–	–	3.8 (0.4)	**3.1**	**4.5**	**−1.0**
Hedonic motivation	Control	7.6 (0.5)	**6.7**	**8.6**	7.6 (0.4)	6.7	8.4	5.2 (0.4)	**4.4**	**6.1**	4.3 (0.5)	**3.3**	**5.3**	**−1.4**
	Prevent	8.2 (0.5)	**7.2**	**9.2**	8.2 (0.5)	7.3	9.1	6.4 (0.5)	5.5	7.3	4.9 (0.5)	**3.9**	**5.9**	**−1.4**
	Substitute	8.3 (0.5)	**7.4**	**9.3**	7.5 (0.4)	6.6	8.3	5.3 (0.4)	**4.5**	**6.2**	5.2 (0.5)	**4.2**	**6.2**	**−1.5**

Note. Significant differences between weeks within-group are highlighted in bold text. There were no statistically significant between-group differences at $p < 0.05$. Habit strength is measured from 1 to 7 and hedonic motivation from 1 to 10, with higher scores representing higher habit strength and greater hedonic motivation for pre-sleep electronic device use.

3.5. Secondary outcomes: acceptability and feasibility

3.5.1. Delivery

Ten participants received email reminders for their daily diary rather than text messages, five due to personal preference and five as text messages could not be sent to participants' country. Fifteen participants (18.5 %) were lost to follow up due to not scheduling their coaching session; the online session was a core part of the program and necessary to continue in Week Two. Completion rates for daily dairy entries in the final sample were high, displayed in Table 4.

3.5.2. Adherence to goals and substitute behaviours

The mean number of days which participants adhered to their goal (Control) or substitute behaviour (Prevent, Substitute) in Week Two of the protocol is shown in Fig. 7. Adherence days were highest in the Control group ($M = 4.3$ days, $SD = 1.9$), followed by Prevent ($M = 3.6$ days, $SD = 1.9$), and Substitute ($M = 3.4$ days, $SD = 1.9$). Non-adherence days were highest in the Prevent group ($M = 2.5$ days, $SD = 1.7$), followed by Substitute ($M = 2.3$ days, $SD = 1.6$) and Control ($M = 1.6$ days, $SD = 1.7$).

3.5.3. Self-monitoring

Self-monitoring was used in Week One, to aid in all participants' development of goals or substitute behaviours. During the coaching sessions, many participants described the self-monitoring questions as insightful (e.g., *"They were soul opening … every single morning I would feel guilty,"* Female, 27, Prevent). As self-monitoring was not linked to motivation as a Mechanism of Action (MoA; [31]), it was also used in the

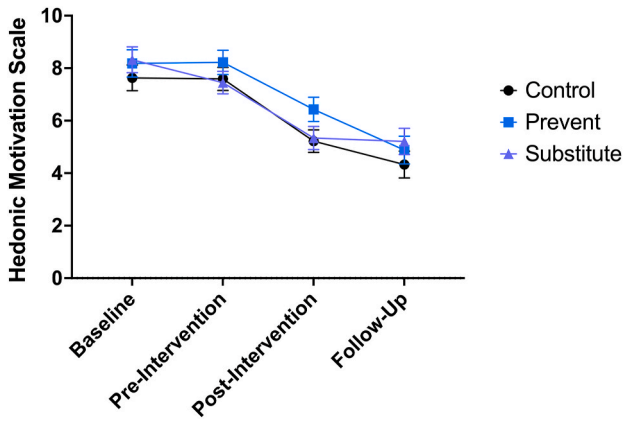


Fig. 6. Mean baseline, pre-post intervention and follow-up changes in hedonic motivation for pre-sleep electronic device use.

Table 4
The percentage of completed daily diary entries across groups in Week 1 and Week 2.

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	*M*
Week 1 (%)	92.7	96.4	87.3	87.3	90.9	87.3	70.9	87.5
Week 2 (%)	96.4	90.9	85.5	81.8	87.3	74.6	76.4	84.7

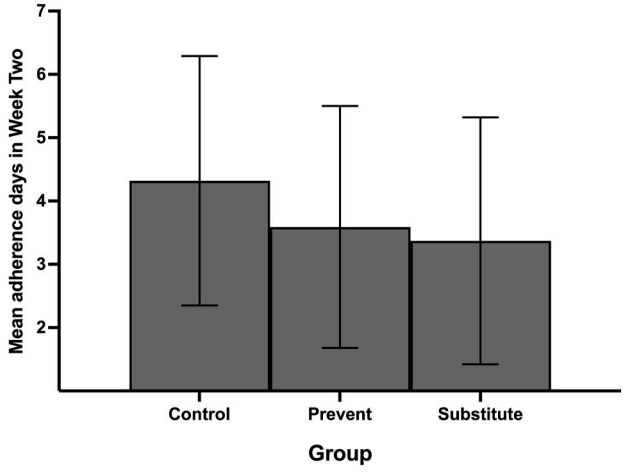


Fig. 7. Mean number of adherence days participants adhered to either their goal or substitute behaviour in Week Two.

Control group. However, many also described self-monitoring as motivating to change their behaviour (e.g., *"Just answering the questions made me more motivated to go to bed earlier,"* Male, 30, Control).

3.5.4. Goal-setting

The Control group were prompted to set goals to address procrastinatory device use (e.g., "Put devices away at 10:30pm"; "Will not use phone after starting nighttime routine"; all goals outlined in Supplemental Table 1). However, it was difficult to ensure the Control group did not use a substitute behaviour instead of setting a goal to abstain from or minimise pre-sleep electronic device use. For example, in the first week of coaching sessions, 3 out of 7 Control participants specifically voiced that they wanted another behaviour to replace their screen time (e.g., *"I need something to replace that winding down,"* Male, 34, Control). Unprompted, many elected their own substitute behaviour and proceeded to incorporate it into their goal (e.g., *"After 11pm I will minimise screen and read a book,"* Male, 26, Control).

3.5.5. Prevent vs. substitute groups

The only difference between the Prevent and Substitute group protocols was the timing of the substitute behaviour: the Prevent group was asked to adopt a substitute behaviour *during the day* to try to prevent bedtime procrastination, while the Substitute group was asked to engage with the substitute behaviour *in the bedtime period.* During the coaching session, many Prevent group participants voiced their inability to find the time to incorporate a substitute behaviour into their day (e.g., *"It would be interruptive at other times of the day"* Male, 23, Prevent). Sometimes, participants would report their job was not conducive to their preferred substitute behaviour (e.g., "*I'm a truck driver, so I don't always get breaks [for me time] anytime I want to*" Male, 32, Prevent). Many participants in the Prevent group elected to adopt a substitute behaviour in the evening period (i.e., before or after dinner) on their own accord.

4. Discussion

The aim of this study was to examine the efficacy and feasibility of the REST-O bedtime procrastination intervention targeting new career starters' pre-sleep electronic device use. Findings indicated initial evidence of behaviour change for electronic device use in the pre-sleep period. The mean pre-post intervention change in pre-sleep electronic device use was 28.7 min per day in the Prevent group (45 % decrease) and 23.1 min per day in the Substitute group (53 % decrease), in support of hypothesis one. There was a small increase in pre-post intervention daily sleep duration in the Prevent and Substitute groups, however, as the increase was non-significant, hypothesis two was not supported. In contrast to both hypotheses one and two, the Control group also experienced reductions in pre-sleep device use and an increase in daily sleep duration. In support of our third hypothesis, weekly measurements revealed that at follow-up, bedtime procrastination and excessive device use decreased, and sleep quality and duration improved compared with baseline. With indication of intervention effectiveness and a high daily completion rate, the results suggest that the program appears to be feasible and acceptable to participants.

Notably, the intervention program resulted in a reduction in bedtime procrastination behaviours in new career starters, extending on previous findings in a general population [24]. While the approach from Jeoung et al. [24] included four 50-min weekly sessions with a therapist, our results indicate that benefits may be achieved through a less resource-intensive automated approach (i.e., one coaching session supplemented with a video and text messages). Conversely, previous goal setting interventions have not seen a decrease in bedtime procrastination [23]. The present findings suggest that the addition of some theory-based BCTS, such as self-monitoring and a short interview or coaching session, could improve the efficacy of future approaches utilising goal setting. Further, results indicate that habit-based strategies may be effective in reducing bedtime procrastination. At follow up, bedtime procrastination and excessive device use continued to decrease, and sleep quality and duration continued to improve in all groups. Although a short follow-up period was utilised, there was no decline back to baseline behaviour as has been seen in interventions based on willpower [21]. As theorised [25] and indicated by formative work ([2], Hill et al., *under review*), interventions grounded in dual-process models are promising approaches to change sleep-related behaviours.

The findings of the present study align with recent research which proposes a bidirectional model of sleep and technology use [20]. While the two experimental groups (Prevent and Substitute) saw a decrease in pre-sleep electronic device use with strategies designed to limit or replace technology use, changes in sleep behaviour may also influence pre-sleep technology use: the Control group showed a decrease in pre-sleep electronic device use by primarily setting goals around sleep.

Further, as technology may be used as a time filler or for emotional regulation prior to sleep [2,20], the reduction in pre-sleep device use may be due to alternative methods to manage time and emotions through substitute behaviours. Our results add to a limited evidence base of interventions designed to reduce electronic device use in adult populations, particularly in the bedtime period [20,45]. Outside of the context of sleep, there is ongoing debate about the effectiveness of interventions designed to ban technology use or promote "digital detoxes" [45], partly due to the complexity of integrating these approaches into daily life [46]. Present findings indicate that adopting a harm minimisation approach towards technology use, rather than outright bans or "digital detoxes", can lead to behaviour change, a reduction in hedonic motivation for device use, and beneficial outcomes for sleep.

The REST-O intervention program appears feasible and acceptable to participants; however, changes to the protocol would strengthen the methodology for a future trial. The active Control group saw unexpected decreases in the target behaviour, which may be attributed to several factors. First, the Control group included self-monitoring and a scaled-back version of the core intervention component, the coaching program, both of which may have induced behaviour change. Second, the Control group participants' personalised goal setting process in some cases prompted substitute behaviour, which may have inadvertently disrupted habitual device use and reduced hedonic motivation. Third, the Control group exhibited the highest adherence rate, suggesting a particularly motivated subset of participants. Fourth, demand characteristics may have influenced responses, particularly given the self-monitoring component. A future program may consider a waitlist control, not using self-monitoring in the Control group, and combining Prevent/Substitute into one experimental group. Finally, non-adherence days were highest in the Prevent group, who expressed difficulty in adopting a substitute behaviour during the day due to busy work schedules – a barrier reflected in the literature [47,48]. New career starters may be better suited to a substitute behaviour in the evening period.

To our knowledge, this intervention program is the first to use a theory-backed, behaviour change approach to reducing bedtime procrastination behaviours. While providing initial indications of efficacy, the present study also has limitations. First, given the pilot nature of this study the sample size may be underpowered for determining statistically significant effects, and a larger sample is necessary to ensure generalisability of the findings. Second, as this study focused on international new career starters, further research is needed to explore its applicability across other, or more localised, adult populations and/or ethnicities. Third, this study did not assess mental health symptoms or psychosocial factors, but future research should include validated mental health measures to examine their role in bedtime procrastination and changes over time post-intervention. Fourth, the study utilised self-report data for pre-sleep electronic device use and sleep measures, which, while providing valuable insights, could be complemented by device-based measures in future studies. Incorporating fatigue measures would help assess the impact of increases in sleep duration on daytime functioning. Additionally, the relatively short follow-up period means long-term changes are unknown. Despite these limitations, the present study lays a solid foundation for future work further exploring the potential benefits of the REST-O intervention.

In conclusion, this study provides initial evidence that a theory-based behaviour change intervention targeting bedtime procrastination can be effective in reducing pre-sleep electronic device use, habit strength and hedonic motivation towards pre-sleep electronic device use, and in improving sleep quality and duration. This intervention represents a promising step in modifying sleep-related health behaviours through personalised behaviour change strategies.

Statement on the use of generative AI

Generative AI services were not used in the preparation of this work.

Trial preregistration

ACTRN12624000105549 (Australian New Zealand Clinical Trials Registry, registered prospectively on 7 February 2024).

CRediT authorship contribution statement

Vanessa M. Hill: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sally A. Ferguson:** Writing – review & editing, Supervision, Conceptualization. **Amanda L. Rebar:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Hailey Meaklim:** Writing – review & editing, Methodology. **Grace E. Vincent:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Data availability statement

The data that support the findings of this study are available from the corresponding author, VH, upon reasonable request.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Vanessa M Hill reports financial support was provided by Fellowships Fund Inc. Vanessa M Hill reports a relationship with Samsung that includes: consulting or advisory. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.sleep.2025.02.043.

References

[1] Kroese FM, De Ridder DT, Evers C, Adriaanse MA. Bedtime procrastination: introducing a new area of procrastination. Front Psychol 2014;5:611. https://doi.org/10.3389/fpsyg.2014.00611.
[2] Hill VM, Ferguson SA, Vincent GE, Rebar AL. 'It's satisfying but destructive': a qualitative study on the experience of bedtime procrastination in new career starters. Br J Health Psychol 2024;29(1):185–203.
[3] Hill VM, Rebar AL, Ferguson SA, Shriane AE, Vincent GE. Go to bed! A systematic review and meta-analysis of bedtime procrastination correlates and sleep outcomes. Sleep Med Rev 2022:101697. https://doi.org/10.1016/j.smrv.2022.101697.
[4] Kroese FM, Evers C, Adriaanse MA, de Ridder DT. Bedtime procrastination: a self-regulation perspective on sleep insufficiency in the general population. J Health Psychol 2016;21(5):853–62. https://doi.org/10.1177/1359105314540014.
[5] Lastella M, Rigney G, Browne M, Sargent C. Electronic device use in bed reduces sleep duration and quality in adults. Sleep Biol Rhythm 2020;18(2):121–9. https://doi.org/10.1007/s41105-019-00251-y.
[6] Hartstein LE, Mathew GM, Reichenberger DA, Rodriguez I, Allen N, Chang A-M, Chaput J-P, Christakis DA, Garrison M, Gooley JJ, Koos JA, Van Den Bulck J, Woods H, Zeitzer JM, Dzierzewski JM, Hale L. The impact of screen use on sleep health across the lifespan: a National Sleep Foundation consensus statement. Sleep Health 2024. https://doi.org/10.1016/j.sleh.2024.05.001.
[7] Pandya A, Lodha P. Social connectedness, excessive screen time during COVID-19 and mental health: a review of current evidence. Frontiers in Human Dynamics 2021;3:684137.
[8] Exelmans L. Electronic Media Use and Sleep: a Self-Control Perspective. Current Sleep Medicine Reports 2019;5(3):135–40. https://doi.org/10.1007/s40675-019-00147-w.

[9] Gradisar M, Wolfson AR, Harvey AG, Hale L, Rosenberg R, Czeisler CA. The sleep and technology use of Americans: findings from the national sleep foundation's 2011 sleep in America poll. J Clin Sleep Med 2013;9(12):1291–9. https://doi.org/10.5664/jcsm.3272.
[10] Rod NH, Dissing AS, Clark A, Gerds TA, Lund R. Overnight smartphone use: a new public health challenge? A novel study design based on high-resolution smartphone data. PLoS One 2018;13(10):e0204811. https://doi.org/10.1371/journal.pone.0204811.
[11] Valasareddy M, Wang W, Abdul-Al CF, Niles SP. The impact of bedtime smartphone usage on sleep health: a pilot quantitative study. Issues in Information Systems 2019;20(4).
[12] Levenson JC, Shensa A, Sidani JE, Colditz JB, Primack BA. Social media use before bed and sleep disturbance among young adults in the United States: a nationally representative study. Sleep 2017;40(9). https://doi.org/10.1093/sleep/zsx113.
[13] Bhat S, Pinto-Zipp G, Upadhyay H, Polos PG. "To sleep, perchance to tweet": in-bed electronic social media use and its associations with insomnia, daytime sleepiness, mood, and sleep duration in adults. Sleep Health 2018;4(2):166–73. https://doi.org/10.1016/j.sleh.2017.12.004.
[14] Exelmans L, Van den Bulck J. Bedtime mobile phone use and sleep in adults. Soc Sci Med 2016;148:93–101. https://doi.org/10.1016/j.socscimed.2015.11.037.
[15] Bartel K, Scheeren R, Gradisar M. Altering adolescents' pre-bedtime phone use to achieve better sleep health. Health Commun 2019;34(4):456–62. https://doi.org/10.1080/10410236.2017.1422099.
[16] Perrault AA, Bayer L, Peuvrier M, Afyouni A, Ghisletta P, Brockmann C, Spiridon M, Hulo Vesely S, Haller DM, Pichon S, Perrig S, Schwartz S, Sterpenich V. Reducing the use of screen electronic devices in the evening is associated with improved sleep and daytime vigilance in adolescents. Sleep 2019;42(9). https://doi.org/10.1093/sleep/zsz125.
[17] Perrault AA, Bayer L, Peuvrier M, Afyouni A, Ghisletta P, Brockmann C, Spiridon M, Vesely SH, Perrig S, Schwartz S, Sterpenich V. Imposing a curfew on the use of screen electronic devices improves sleep and daytime vigilance in adolescents. Cold Spring Harbor Laboratory; 2018.
[18] Das-Friebel A, Perkinson-Gloor N, Brand S, Dewald-Kaufmann JF, Grob A, Wolke D, Lemola S. A pilot cluster-randomised study to increase sleep duration by decreasing electronic media use at night and caffeine consumption in adolescents. Sleep Med 2019;60:109–15. https://doi.org/10.1016/j.sleep.2018.11.010.
[19] Flint Bretler O, Tzischinsky O, Asraf K, Shochat T. The effects of parental intervention on sleep patterns and electronic media exposure in young adolescents. Clocks & Sleep 2022;4(1):129–44. https://doi.org/10.3390/clockssleep4010013.
[20] Bauducco S, Pillion M, Bartel K, Reynolds C, Kahn M, Gradisar M. A bidirectional model of sleep and technology use: a theoretical review of How much, for whom, and which mechanisms. Sleep Med Rev 2024;76:101933. https://doi.org/10.1016/j.smrv.2024.101933.
[21] Wood W, Neal DT. Healthy through habit: interventions for initiating & maintaining health behavior change. Behavioral Science & Policy 2016;2(1):71–83. https://doi.org/10.1177/237946151600200109.
[22] Evans J, Stanovich KE. Dual-process theories of higher cognition: advancing the debate. Perspect Psychol Sci 2013;8(3):223–41. http://www.jstor.org.ezproxy.cqu.edu.au/stable/44289870.
[23] Valshtein TJ, Oettingen G, Gollwitzer PM. Using mental contrasting with implementation intentions to reduce bedtime procrastination: two randomised trials. Psychol Health 2020;35(3):275–301. https://doi.org/10.1080/08870446.2019.1652753 [Article].
[24] Jeoung S, Jeon H, Yang HC, An H, Suh S. A randomized controlled trial of a behavioral intervention for decreasing bedtime procrastination using a wait-list control group in a non-clinical sample of young adults. Sleep Med 2023;108:114–23. https://doi.org/10.1016/j.sleep.2023.06.001.
[25] Rebar AL, Reynolds AC, Ferguson SA, Gardner B. Accounting for automatic processes in sleep health. J Sleep Res 2020;29(5). https://doi.org/10.1111/jsr.12987.
[26] Williams DM. The theory of hedonic motivation. In: Williams DM, editor. Darwinian hedonism and the epidemic of unhealthy behavior. Cambridge University Press; 2019. p. 139–46. https://doi.org/10.1017/9781316275047.017.
[27] Michie S, Richardson M, Johnston M, Abraham C, Francis J, Hardeman W, Eccles MP, Cane J, Wood CE. The behavior change Technique taxonomy (v1) of 93 hierarchically clustered techniques: building an international consensus for the reporting of behavior change interventions. Ann Behav Med 2013;46(1):81–95. https://doi.org/10.1007/s12160-013-9486-6.
[28] Gardner B, Richards R, Lally P, Rebar A, Thwaite T, Beeken RJ. Breaking habits or breaking habitual behaviours? Old habits as a neglected factor in weight loss maintenance. Appetite 2021;162:105183. https://doi.org/10.1016/j.appet.2021.105183.
[29] Hjetland GJ, Skogen JC, Hysing M, Sivertsen B. The association between self-reported screen time, social media addiction, and sleep among Norwegian university students. Front Public Health 2021;9:794307. https://doi.org/10.3389/fpubh.2021.794307.
[30] Dai H, Milkman KL, Riis J. The fresh start effect: temporal landmarks motivate aspirational behavior. Manag Sci 2014;60(10):2563–82. https://doi.org/10.1287/mnsc.2014.1901.
[31] Johnston M, Carey RN, Connell Bohlen LE, Johnston DW, Rothman AJ, de Bruin M, Kelly MP, Groarke H, Michie S. Development of an online tool for linking behavior change techniques and mechanisms of action based on triangulation of findings from literature synthesis and expert consensus. Transl Behav Med 2021;11(5):1049–65. https://doi.org/10.1093/tbm/ibaa050.
[32] Zoom. (Version 5.17.11) Zoom video communications. https://www.zoom.us/; 2024.
[33] Kang M, Ragan BG, Park JH. Issues in outcomes research: an overview of randomization techniques for clinical trials. Journal of athletic training 2008;43(2):215–21.
[34] Sancho-Domingo C, Carballo JL, Coloma-Carmona A, Buysse DJ. Brief version of the Pittsburgh Sleep Quality Index (B-PSQI) and measurement invariance across gender and age in a population-based sample. Psychol Assess 2021;33:111–21. https://doi.org/10.1037/pas0000959.
[35] Gardner B, Abraham C, Lally P, De Bruijn G-J. Towards parsimony in habit measurement: testing the convergent and predictive validity of an automaticity subscale of the Self-Report Habit Index. Int J Behav Nutr Phys Activ 2012;9(1):102. https://doi.org/10.1186/1479-5868-9-102.
[36] Kwon M, Lee JY, Won WY, Park JW, Min JA, Hahn C, Gu X, Choi JH, Kim DJ. Development and validation of a smartphone addiction scale (SAS). PloS one 2013;8(2):e56936.
[37] Eisinga R, Grotenhuis Mt, Pelzer B. The reliability of a two-item scale: pearson, Cronbach, or Spearman-Brown? Int J Publ Health 2013;58:637–42.
[38] Bastien CH, Vallières A, Morin CM. Validation of the Insomnia Severity Index as an outcome measure for insomnia research. Sleep Med 2001;2(4):297–307.
[39] Vincent GE, Aisbett B, Hall SJ, Ferguson SA. Fighting fire and fatigue: sleep quantity and quality during multi-day wildfire suppression. Ergonomics 2016;59(7):932–40.
[40] R Core Team. _R: a language and environment for statistical computing_. R Foundation for Statistical Computing; 2023. https://www.R-project.org/.
[41] Bates D, Mächler M, Bolker B, Walker S. Fitting linear mixed-effects models using lme4. 2015. arXiv preprint arXiv:1406.5823.
[42] Lenth R. _emmeans: estimated marginal means, aka least-squares means_. https://CRAN.R-project.org/package=emmeans; 2023.
[43] Cohen J. *Statistical power analysis for the behavioral sciences*. Routledge; 1988.
[44] Schulz KF, Altman DG, Moher D. CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials. J Pharmacol Pharmacother 2010;1(2):100–7. https://doi.org/10.4103/0976-500X.72352.
[45] Nassen L-M, Vandebosch H, Poels K, Karsay K. Opt-out, abstain, unplug. A systematic review of the voluntary digital disconnection literature. Telematics Inf 2023;81. https://doi.org/10.1016/j.tele.2023.101980.
[46] Radtke T, Apel T, Schenkel K, Keller J, Von Lindern E. Digital detox: an effective solution in the smartphone era? A systematic literature review. Mobile Media & Communication 2022;10(2):190–215. https://doi.org/10.1177/20501579211028647.
[47] Paterson JL, Reynolds AC, Duncan M, Vandelanotte C, Ferguson SA. Barriers and enablers to modifying sleep behavior in adolescents and young adults: a qualitative investigation. Behav Sleep Med 2019;17(1):1–11. https://doi.org/10.1080/15402002.2016.1266489.
[48] Oliver M, Rodham K, Taylor J, Mciver C. Understanding the psychological and social influences on office workers taking breaks; a thematic analysis. Psychol Health 2021;36(3):351–66. https://doi.org/10.1080/08870446.2020.1764954.